

Experience in applying OR techniques to the solution of practical resource management problems

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This paper gives a brief overview of some successful applications of operations research techniques to resource management. It covers forecasting and scheduling, the Work Manager System, simulation modelling and development of a 'tour-build' algorithms.

1. Introduction

This paper considers the experience of BT's Management Science Consultancy Unit (MSCU) in applying operations research (OR) techniques to a number of resource management problems. The paper begins with a discussion of forecasting and scheduling customer service operations, followed by sections on the management of engineers through the 'real time' work manager system and the use of simulation modelling to support policy and decision making in this area. The final part of the paper describes some earlier work on developing 'tour-build' algorithms for allocating work to the field. In each case, the developments are now in operational use within BT.

2. Customer service centres

BT's customer service centres (CSCs) are responsible for handling sales and billing enquiries, and fault reports. The day-to-day operational problem faced by a CSC manager is how best to deploy the flexible, but finite, people resource in order to meet quality of service targets.

This problem has three distinct elements:

- how to predict the amount of work that will arrive at a centre in each period of the day,
- how to convert this prediction into a forecast of the number of people needed to handle the work,
- how to produce a set of attendances that match the people available to the predicted work.

A prototype tool has been developed which tackles the above forecasting problems.

2.1 Forecasting work

The data available on which to base a forecast of calls is potentially extensive. The automatic call distribution equipment which allocates calls to servers can supply data by day and period. A study of such data revealed that:

- there was a regular pattern of calls (see Fig 1),

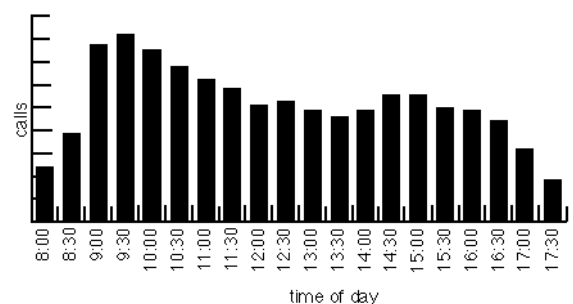


Fig 1 Typical pattern of calls within a day.

- there were differences between days of the week (see Fig 2),

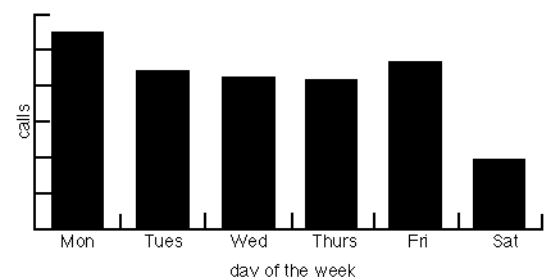


Fig 2 Typical pattern of calls by day of week.

- there were no clearly defined monthly, quarterly or seasonal patterns.

It was also observed that the pattern of calls within the day was relatively stable whatever the number of calls in the day.

It was therefore concluded that an appropriate forecasting method was as follows:

- forecast, for each day of the week separately, the total number of calls in the day using an ARIMA [1] model fitted to up to two years data on the total number of calls in the day,
- estimate the number of calls within a period of the day by calculating the average proportion of the day's calls that arrived in the period using data from the most recent 13 weeks.

Bank holidays, days close to bank holidays and outliers in the data series were handled as special cases.

Tests were carried out, on historic CSC data, to see if the results produced were better than simple alternatives (e.g. assuming tomorrow will be the same as yesterday). The chosen method produced, on average, the most accurate results of the methods tested. Other methods that were considered but rejected were:

- Harrison-Stevens Bayesian [2],
- exponential smoothing [3],
- Holt-Winter's [3].

Bayesian methods were rejected on the grounds that:

- they are too complex to program and no suitable software was available,
- although Bayesian methods allow the incorporation of subjective user knowledge to improve forecasts, there was little useful knowledge available.

Holt-Winter's method and exponential smoothing were rejected on the grounds that:

- it was thought that they could be improved upon in terms of accuracy,
- other methods offered more scope for later development.

2.2 Predicting people required

Queuing theory is used to convert forecasts of call numbers to numbers of servers needed [4]. For a bounded queue with s servers and offered traffic ρ the wait time distribution is given by:

$$P(\text{wait} < t) = P_s \frac{\exp(-t(s - \rho))}{1 - \frac{\rho}{s}}$$

where the probability P_s that the system has s calls in it (i.e. all servers are busy and no-one is waiting) is given by:

$$P_s = \frac{(s - \rho)E_s(\rho)}{s - \rho + \rho E_s(\rho)}$$

and $E_s(\rho)$ is the Erlang loss function, defined by:

$$E_s(\rho) = \frac{\frac{\rho^s}{s!}}{\sum_{v=0}^s \frac{\rho^v}{v!}}$$

For large s and ρ this can be evaluated by the approximation:

$$E_s \rho = \frac{\exp\left\{s \ln\left(\frac{\rho}{s+1}\right) - \rho + s + 1 - \frac{1}{12(s+1)} + \frac{1}{360(s+1)}\right\}}{Q(x) \sqrt{2\pi(s+1)}}$$

with Q of the form:

$$Q(y) = 0.5\{1 + y(d_1 + y(d_2) + y(d_3) + y(d_4 + y(d_5 + y(d_6))))\}^{-16}$$

for specific constants d_1 to d_6

and:

$$x = z - \frac{\left\{ \frac{2(z^2 - 1)}{3\sqrt{s}} - \frac{z(z^2 - 3)}{4} \right\}}{27s}$$

$$z = 3\sqrt{s+1} \left\{ 3\sqrt{1 + \frac{\rho}{s}} + \frac{1}{9(s+1)} - 1 \right\}$$

The above provides a measure for calculating the percentage of calls answered (PCA) knowing the call arrival rate, the number of servers and handling time. What is required is to compute the number of servers required. A simple root finder incorporating the above formulae is used for this purpose. A further refinement is that the PCA calculation is carried out a number of times with perturbed values of the call arrival rate, around the predicted value, in order to reflect the uncertainty associated with the forecast value.

2.3 Scheduling attendances

Following an investigation of the various methods available for scheduling attendances, it was decided to use a heuristic approach. Experience has shown that although the attendances produced by the algorithm are not necessarily optimal, they are a good fit. Extensive benchmarking of the algorithm has not yet taken place but a comparison with proprietary systems suggests that these results are broadly similar.

3. Work management

The task of assigning jobs to engineers is performed by Work Manager [5] which is based on a real time algorithm designed to allocate one job at a time to engineers taking account of travel time, the skills required and the need to meet business targets. The core of the algorithm is based upon the 'Hungarian' solution [1] to the general assignment problem.

In its most basic form, the work assignment problem consists of n engineers and n jobs together with a numerical score or cost for processing the work (see Fig 3). The 'assignment problem' is the quest for an assignment of a single engineer to one and only one job in a manner that minimises the sum of the numerical scores or costs. Solving this problem is non-trivial, for example a 10×10 matrix has 3628 800 possible assignments; increasing to an 11×11 matrix gives 39 916 800.

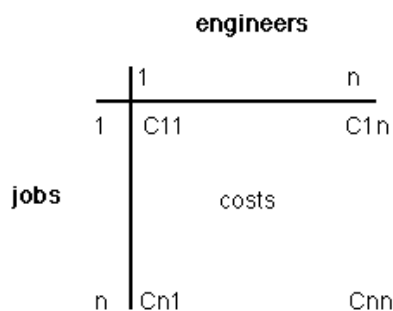


Fig 3 Allocation matrix.

This algorithm employs 'pre-allocation' to change the information on which assignment is based. The calculations are updated every 5 minutes or less.

There are two steps in the allocation process:

- determining which jobs are included in the matrix,
- determining which jobs will be allocated next and to which engineer.

3.1 Which jobs to include in the matrix

One of the critical factors here is the size of the matrix in relation to the number of engineers available. As the matrix size increases, the time to find a minimum cost

assignment increases, approximately as a cubic power function. In order to keep this time to a manageable level:

- the maximum number of jobs is restricted to 300,
- problems can also arise if the matrix size is larger than the number of jobs in the system.

Each job in the system has an importance score which reflects its absolute priority, independent of its business target time. Factors taken into account include the job type, appointed or not, and emergency requirements.

In deciding which jobs to include in the matrix, the system calculates a priority score as follows:

$$\text{Priority score} = \text{Importance score} - k_1 * [\text{business target time} - (\text{current time} + \text{job duration})]$$

k_1 is a parameter used to fine-tune the priority rating.

3.2 Which jobs will be allocated next and to which engineer

The next step is deciding (provisionally) which job will be given to each engineer when they next access the system. The aim is to minimise travel and the number/ extent of failures and to make good use of the available skill set.

This is done by calculating a 'cost score' for each job for each engineer which reflects the travel time and by how much (if at all) the job would fail if it were allocated to that person. The aim then is to find the combination of jobs and engineers which minimises the total of this score. The general form of the cost score is given in section 3.3.

With more than one job per engineer jobs can appear in the matrix but not be pre-assigned to a real person. In this situation, it can be said that jobs have been allocated to a 'dummy engineer', analogous to delaying their allocation until later. The effect is that the minimum cost solution is to allocate to 'dummy engineers' those jobs which, if they were allocated, would have the highest score. There are two reasons a job might have a high score for all engineers:

- it is remote, in which case the best course may be to leave it until an engineer is in the area,
- it has already failed.

Assigning these latter jobs to 'dummy engineers' is effectively concentrating on 'winners'. The downside of this is that failed jobs might never get cleared. This potential problem is avoided by forcing jobs to be preassigned to real engineers after a certain time.

The 'duration to clear time' (DCT) parameter which controls this forced assignment has values for different

categories of jobs. Work manager can define up to four different categories of job — target, focus, core and basic. These form a hierarchy in which target is used for jobs with high importance scores and basic for jobs with low importance scores. DCT values can then be set to meet any differing business requirements on these jobs.

3.3 The form of cost score

The algorithm identifies the combination of jobs and engineers which minimises the following cost score summed across all technicians:

$$\text{failure cost} + k_2 \times \text{travel time} \\ [\dots + \text{overtime/idle time effects}]$$

where:

$$\begin{aligned} \text{failure cost} &= \text{importance score of job} \\ &\times P(\text{fail}) \text{ if } P(\text{fail}) < 1 \\ &\quad \text{importance score of job} \\ &\times (1 + \text{cfr} \times \text{minimum time after failure}) \\ &\quad \text{if } P(\text{fail}) = 1 \end{aligned}$$

$P(\text{fail})$ = probability the job will fail if allocated to that engineer based on their earliest and latest time for completing their current job, the travel time to the next job and the time to complete that job (for jobs which need to be completed by their business target time),

cfr = cost of failure rate parameter,

k_2 = travel time/job failure balance parameter.

It should be noted that appointment jobs which have failed are dropped from the matrix and have to be handled manually. For appointments there is an additional element in the cost score to cover any risk of arriving early for appointments. These are classed as failures and can lead to wasted time. This additional cost is calculated as:

$$k_2 \times k_3 \times \text{expected idle time before the start of the appointment}$$

where k_3 = appointment early time penalty parameter.

4. Decision support

Discrete event simulation of field operations has been used to determine a range of policy issues, including appointment setting and multiskilling. This work has resulted in the development of the field force job allocation model (FFJAM), which allows different field scenarios to be assessed against parameters such as productivity and

quality of customer service to help determine BT field practices (e.g. appointment slots offered to customers, levels of engineer multiskilling). FFJAM simulates the allocation of reactive repair jobs to engineers as performed in the field by the Work Manager real time algorithm. It allows different scenarios to be assessed against a variety of performance indicators, including productivity and quality of service.

4.1 Using the model

FFJAM handles jobs which have:

- a specific appointment,
- clear times based on internal business targets,
- a response time based on specific contractual arrangements.

To use the model two data files are constructed: one with engineer details (e.g. skills, start and finish locations, start and finish times), and the other with job details (e.g. location, type of job, commitment time).

The results from FFJAM show the impact of specific scenarios (compared to current working practices) on productivity and quality of service (see Fig 4).

A tracer facility is also available which traces the route patterns of engineers throughout the day (see Fig 5).

4.2 Job allocation

In deciding which job to give an engineer, FFJAM will take account of both travel time and the need to meet customer commitments. The trade-off between travel time and failures will be determined by the user, allowing, for example, a high emphasis on minimising travel in overload conditions. Similarly the relative importance of different types of job will be user defined.

4.3 Engineer travel

FFJAM takes account of the variability between the estimated travel time (provided by the Work Manager real time algorithm — see section 3) for travel to and from jobs, and actual travel time. This variability can be caused by, for example, poor road traffic flows. The best fit distribution for actual travel time, given the estimated travel time, was the Lognormal (see Fig 6).

A random sample of numbers was generated from the Lognormal distribution, with parameters m and s , from the standard normal distribution using the following formula.

$$L:m,s \sim \text{mexp}(sN;0,1)$$



Fig 4 Example of FFJAM analysis.

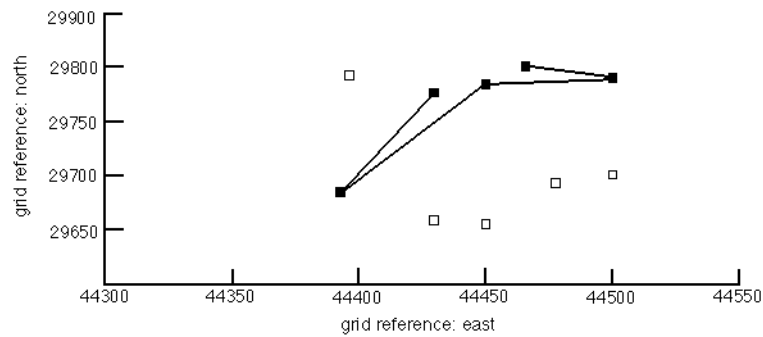


Fig 5 Tracing the engineers' job tours.

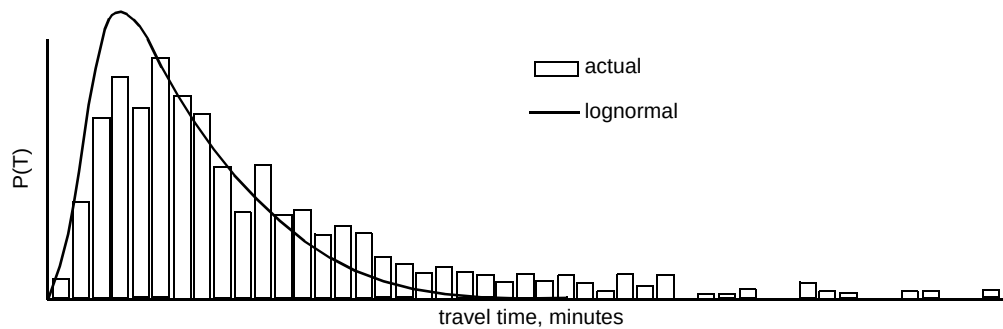


Fig 6 Distribution of actual travel time for given estimated time.

The relationships between m , s and μ , σ , where μ , σ are the mean and standard deviation of the Lognormal distributions, are given below:

To generate a random number from a standard normal distribution the following formula was used:

$$N(0,1) \approx \sum_{i=1}^{12} R_i - 6$$

$$m^2 = \frac{\mu^4}{\mu^2 + \sigma^2}$$

$$s^2 = 1n \left(1 + \frac{\sigma^2}{\mu^2} \right)$$

where R_j is a random number between 0 and 1.

A simple algorithm [7] was used to produce random numbers in the range 0-1.

4.4 Simulation modelling approach

Two common discrete simulation modelling approaches were considered for FFJAM:

- event scheduling,
- activity scanning.

Event scheduling

An event-based model is made up of a set of event routines, each of which describes the operations in which entities engage when the system changes state. They describe the event logic and will:

- update all variables affected by the event,
- schedule any unconditional events,
- check and schedule conditional events.

During the operation of an event-scheduling model, the earliest event will be selected by the executive from the event list and executed. This event will then be removed and new events will be scheduled (see Fig 7).

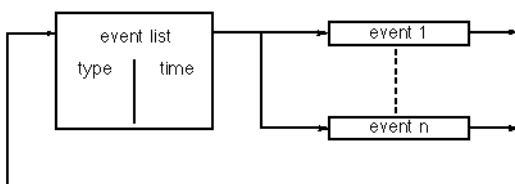


Fig 7 Simulation of event-scheduling approach.

Activity scanning

This approach concentrates on the iterations of entities which perform activities, rather than mapping out operations which follow from a state change, as in the event approach.

Each engineer is assigned a 'clock' which, when the activity starts, is advanced to the activity completion time. If t represents simulated time, and $t(i)$ engineer (i)'s 'clock', then the following tests can be made:

- if $t(i) > t$ then engineer (i) is busy,
- if $t(i) < t$ then engineer (i) is idle,

- if $t(i) = t$ then engineer (i) has just finished current/ last activity.

Each activity is made up of two parts:

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- test — the conditions which must be satisfied if the activity is to be executed,
- actions — the operations of the activity (only executed if tests are passed),

and represents an independent segment of the simulation which will be executed if the tests are passed. At the end of each activity simulated time is advanced to the next event (as indicated by the earliest 'clock' time) by the executive. At each event, each activity is attempted in turn. The actions to be performed at each event then depend on current simulation conditions (see Fig 8).

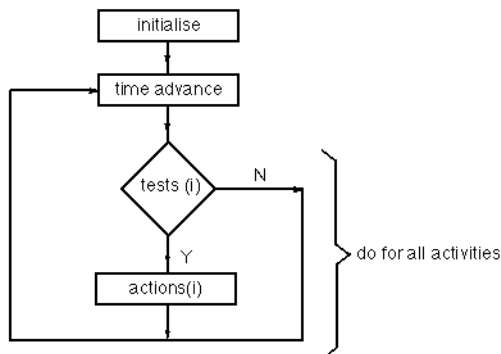


Fig 8 Simulation of activity-scanning approach.

Hence, for the activity-scanning approach the relationship between activities is defined by the present conditions within the simulation, whereas the event scheduling model would require all possible outcomes from each event to be coded.

However, as the activity scanning approach deliberately treats each activity as independent it will scan every activity in turn — even if the simulation conditions mean that only one activity is possible — giving run time inefficiency. An event-based approach, however, will only execute those events known to be possible (as they have been identified and passed to the event scheduler).

4.5 Discussion

The activity scanning approach was selected, in preference to an event scheduling approach, for the development of FFJAM for the following reasons.

- Activity scanning uses small discrete activity modules, and this approach is well suited to a structured programming methodology and top-down design. The developer does not have to be too concerned about the sequence of activities at each event, as this is taken care of by the executive in the activity scan. This makes testing and future maintenance significantly easier.

- It is simpler to program an activity-based approach. Consequently, it is cheaper and quicker in terms of development costs. Although activity scanning runs less efficiently than the event-scheduling approach, the modern computing capabilities reduce run times to an acceptable level.

FFJAM has been used to evaluate a number of proposals for change to working methods. It is currently being used to assess the appropriate level of multitasking to be put in place to enable work areas to cope with their work-loads with fewer engineers.

5. TIMES-WEST

The work described above focused particularly on the problems of 'real time' allocation of field repair work. Prior to Work Manager, the problem of scheduling batches of jobs to engineers was already being investigated. In 1986 there were about 40 000 engineers in BT installing phones and extensions in response to requests logged on the customer service system (CSS). Distribution officers (DOs) were responsible for a team of approximately 12 engineers. With 50 to 80 jobs to allocate this was a time-consuming and non-trivial task impacting on operational effectiveness.

There were two main problem areas.

- There was a requirement to minimise travelling for many engineers simultaneously as well as deal with the many constraints of appointments, skill levels, problem is far more difficult than the travelling salesman problem and there was no existing solution.
- Achievement of the economies required a significant change in the resource management culture. In particular, the TIMES-WEST system introduced the idea of allocating the available work to the fewest number of engineers who could do it rather than distribute the work more or less equally to those available on a day. This freed a number of engineers in each team for a whole or half day for other tasks.

5.1 The design philosophy

The development of the TIMES-WEST system was evolutionary. The initial objective was to create a tool for distribution officers to use to explore the implications of their decisions and to consider alternatives. It would be a companion to CSS, using CSS data and the same terminal. Before any of this could be contemplated, however, the difficult core problem of organising the work had to be solved. There were two observations that were instrumental in starting the design of the algorithms. The first came from the MSCU team leader's experience of beekeeping. On a hot summer's day, when bees leave a hive they come out of

the entrance and immediately take off on a low trajectory. Arriving bees tend to hover over the entrance land above it and enter from above. Under low traffic conditions they can even fly right in. On some afternoons, however, there can be a large number of young bees hovering in a dense cloud around the entrance while other bees leave and arrive navigating their way through the cloud. They achieve this without mid-air collisions. It seemed amazing that they could compute their trajectory under such conditions. But perhaps they did not. Perhaps they only saw a small part of the situation, and used simple strategies which in aggregate seemed complex. Similarly, simple effective heuristics might be found for scheduling the jobs.

The second idea was that there was no need to find an optimum. With these combinatorial problems, finding an optimum solution can take a very long time (or even be impossible within a sensible time) but sub-optimal solutions, say 97% of the optimal, can be found in only a tiny fraction of that time. So long as the solutions generated are usefully better than current practice, the process of automating the work allocation procedure could be worthwhile.

5.2 Facilities

The system has two main components.

- TIMES is a work-scheduling algorithm which programmes daily 'tours' of work, taking account of job skill requirements while minimising travel time. The total work-load is met with the minimum number of tours.
- WEST is an electronic diary system for non-appointment work, in particular for jobs taking more than one day to complete. It provides continuous monitoring to warn of likely jeopardy conditions and has a facility to redeploy engineers at short notice. Availability can be planned for up to three months ahead.

The system operates by taking a download of CSS work queues each night for preprocessing by the TIMES algorithm. Work allocated by TIMES is posted into the WEST diary for monitoring.

The TIMES-WEST system has enabled BT to run the whole of the London business installation work from only two sites with fewer DOs and engineers.

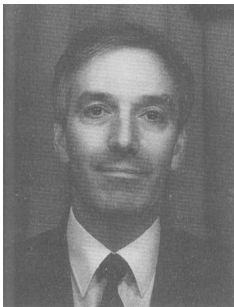
6. Conclusions

Effective resource management is key to improving service to customers while cutting costs. The use of modelling and OR techniques can greatly assist in this but the practitioner needs to be strongly focused on the customer requirements and work closely with the service department

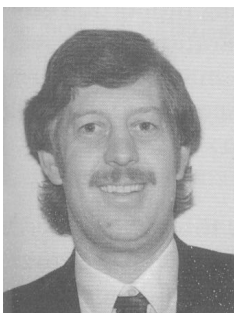
in developing and tailoring techniques to meet their particular requirements. It is necessary to be flexible in the use of both standard and non-standard techniques with the emphasis on improving the current situation. Although BT's operations are large and often complex, simple models and techniques can still yield great benefits

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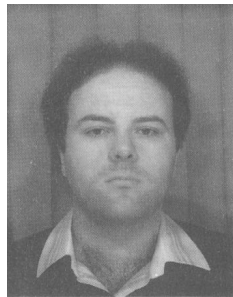
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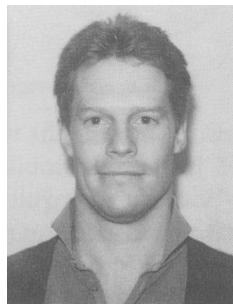
Andy Noble graduated from the University of Newcastle-upon-Tyne in 1976 with a BSc in Statistics. He then joined BT where he worked on manpower planning and depreciation studies. Since then he has worked on many Management Science projects. In addition, he has spent time as a support manager for the roll-out of BT's Total Quality Management programme. He is now in BT's Management Science Consultancy Unit where he is working on the measurement and reporting of BT's quality of service and external customer satisfaction.



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David Morton has a BSc in Applied Statistics and an MSc in Statistics and Operational Research. He joined BT in 1983 and worked initially on producing statistical and investment appraisal software for managers. He followed this with involvement in assessing the effectiveness of BT advertising campaigns and since 1990 has worked on resource allocation problems and in particular Work Management.



Tim Richards received an MSc in Operational Research from Sussex University in 1987, and went on to design Expert System applications in Canada. He joined BT in 1988 concentrating on the practical application of both "hard" and "soft" management science techniques to commercial business issues for BT.



Marcus Burstin graduated with an honours degree in physics from Manchester University in 1960. He received his MSc in OR from London University in 1964 and then joined BT in 1969. He has been in the Management Science consultancy Unit since its inception and is particularly interested in manpower scheduling and service systems. Last Updated : 27-Apr-95